Enhancing ****'s Health Platform with Contactless BP, SpO_2, and BMI Estimation

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1. Background

In recent years, there has been a growing demand for personalized healthcare recommendations to enhance quality of life, which is defined based on numerous aspects, such as an individual's physical and emotional health as well as self-awareness. Traditional devices that rely on photoplethysmography (PPG), such as pulse oximeters, can provide useful biometric insights, but are limited in practicality or capability in holistic understanding. To address these challenges, ****, an AI-driven platform, has introduced ****, designed to deliver personalized advice and enhance individuals' quality of life by measuring ****'s proprietary Life Quality Index (LQi) [1]. To achieve this, **** employs a multi-modal approach to collect health biomarkers such as heart rate, respiratory rate, and stress indicators. Unlike traditional devices, **** introduces a unique feature called Face Scan, powered by remote PPG (rPPG). This innovation allows users to simply use their smartphones to record a short facial video of themselves without any physical contact. Then, **** analyzes to predict the multiple biomarkers which can be used to provide personalized health suggestions. This project aims to expand the range of physical biomarkers supported by the **** platform by investigating the feasibility of predicting blood pressure (BP), blood oxygen (SpO_2), and body mass index (BMI) using facial videos and leveraging deep learning (DL) models, toward clinical standard.

Goal - Expanding Face Scan

The current Face Scan technology is limited in scope, only capturing a narrow set of biomarkers from facial video. By expanding this range, **** is able to generate a more holistic assessment and provide more accurate recommendations to its users. To address this challenge, the system should leverage DL models which have the ability to extract nuanced and complex features from facial video data to predict more biomarkers reliably. The three specific biomarkers to add are **1) BP**, **2) SpO_2**, and **3) BMI** since they are strong indicators of multimorbidity risk [2]. Blood pressure and blood oxygen saturation can be measured indirectly through a facial video by looking at the colour change of the face, which is typically not perceptible by the human eye, but is detectable by cameras. The colour change is caused by blood flowing through the arteries, veins, and capillaries, and the environmental light being absorbed and reflected differently depending on blood flow. BMI only requires a single frame of facial video to do accurate prediction.

Requirements and Deliverables

The following requirements are used to evaluate the project's success:

1. Predictive models must achieve certain accuracy and errors, mean absolute error (MAE) and/or root mean square error (RMSE) on test datasets that must be close to state-of-the-art.
    a. BP < 10 mmHg

b. SpO_2 < 1.5 RMSE
c. BMI - No specific clinical standard so as close to '100% correct' as possible referring to the client

2. A strong Pearson correlation coefficient (PCC) between prediction and ground truth, close to 1.
3. The model should be feasible to integrate into the **** platform.
    a. This means the pre-trained model weights will be provided for inference along with a sample script demonstrating the inference process.
    b. The model code will also be provided for ability to retrain or fine-tune on more data.

2. Data Analysis

Our models are trained on datasets from public sources that are freely available, since our client is not able to provide us their private user data with paired facial videos and ground-truth physiological measurements. We use MCD-rPPG [3], an open, comprehensive dataset with 3,600 facial videos, each approximately 3 minutes long, 24 or 30 frames per second (fps), from 600 subjects and from 3 different camera angles (front, left and right) in 2 states (before and after 15 squats in 30 seconds). Each video is paired with synchronized 100 Hz PPG signals and 13 biomarker measurements, including BP, SpO_2 and BMI, which are measured prior to the recording session with medical equipment: tonometer AND UA-911BT-C for BP, pulse oximeter Beurer PO 60 for SpO_2 and scales REKAM BS 630FT for BMI. These measurements are treated as ground-truth labels for each video. Only the head-on camera angle (FullHD_webcam) is being used, as this is what is expected for the face scan videos. Additionally, only the pre-exercise videos are used since the post-exercise stabilization time varies from person to person, and only one ground truth value is provided for BP and SpO_2. Due to the emerging nature of BP and SpO_2 prediction, many datasets are small and biased towards skin tones (Fig. 9.1.1). By acknowledging that the MCD-rPPG dataset is biased toward white participants as reflected in the race distribution, generated by DeepFace's face recognition model, models trained or tuned on this dataset should not be directly applied to all racial groups and should be fine-tuned using a more diverse dataset. Furthermore, the ground-truth measurements for the 3 biomarkers are skewed and imbalanced. To mitigate the effects of imbalanced distribution, weight strategies are used during training. Approaches such as PyTorch's WeightedRandomSampler or scikit-learn's compute_sample_weight sample minority data more often or assign higher weights to the minority samples in the loss function during DL backpropagation.

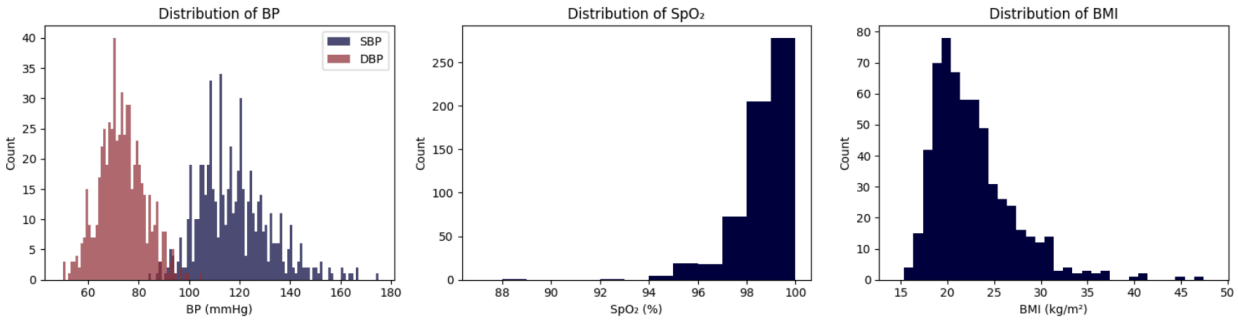


Figure 2.1. Systolic BP (SBP) has a mean of 112.45 mmHg (std 17.43) and diastolic BP (DBP) has a mean of 73.79 mmHg (std 9.25). SpO_2 has a mean of 98.01% (std 1.29) with left skewness of -2.49. BMI has a mean of 22.73 kg/m² (std 4.34) with right skewness of 1.53.

2.1 Blood Pressure Data Processing

For BP data processing, each video is being passed through the client provided DL-based rPPG extraction model, MTTS-CAN [4], which produces face-extracted frames and a normalized blood volume pulse (BVP) by capturing subtle colour changes on the face. It is the same model the client uses for heart rate estimation, so the BP pipeline remains fully compatible with the client's existing infrastructure. Then, heart rate is estimated from pulse signal using the dominant spectral peak. Using this estimated heart rate, the signal is segmented into windows that contain 7 beats, roughly 1 second per beat, to ensure no cardiac cycle is cut in half. Prior work shows that using 7-beat windows minimizes the prediction error compared with other window length [5]. Next, each window is upsampled to 125 Hz to match the input requirements (N×875×1 where N is the number of video segments) of the BP prediction models used later for fine-tuning. The total number of processed segments is 17,840. Lastly, all segments derived from a given video are labeled with corresponding measured ground-truth BP value. For fine-tuning, the post-process dataset is being randomly split into train, validation and test with ratio of 30/10/60.

2.2 Blood Oxygen Data Processing

For SpO_2, the preprocessing converts the videos into spatio-temporal maps (ST-maps). Similar to BP, the faces in the videos are cropped to a region-of-interest (ROI) (area under eyes to the chin). Google's Mediapipe Face landmark model [6] is used to detect the face region. The ROIs are downsampled to 16x14 resolution. ROIs are grouped together for each 10 second segment and flattened vertically to produce a ST-map. The resulting ST-map has dimension NxTxC of 224×300×3 for regular 30 fps videos, and 224×240×3 for 24 fps videos, where N is the flattened ROI, T is the number of frames per 10 second segment, and 3 is the RGB colour channels. The 24 fps ST-maps are interpolated to produce consistent sized ST-maps of 224×300×3. 10 video segments from each subject are taken. The resulting 6000 ST-maps are split into train-val-test split with ratio 75/10/15 for model training and evaluation.

2.3 Body mass index data processing

For BMI, the data is preprocessed using a robust face-detection and alignment model, MTCNN [7], to crop the facial region from the raw video frames. Each detected face is then resized to 224×224 pixels and normalized to ensure consistency with the input requirements of the feature extraction convolutional neural network architectures, like ResNet. These preprocessing steps reduce background noise, standardize the geometry of the input images, and enhance the stability of feature extraction across subjects. To assess model generalization, the dataset is partitioned into training and testing subsets with a 4:1 ratio under a 5-fold cross-validation scheme [8]. This ensures that every subject appears in both training and testing across folds, enabling a more reliable evaluation of performance on this dataset.

3. Methods

3.1 Blood Pressure Methodology

To estimate SBP and DBP from facial videos, the processed rPPG signals are fed into three well-known DL model families commonly used for physiological signal regression: 1D-AlexNet, 1D-ResNet, and 1D-CNN+LSTM [9]. This selection aims to compare spatial CNN-based transfer learning against combinations with temporal sequence-modeling approaches. AlexNet provides a baseline with strong generalization on smaller datasets; ResNet offers deeper feature extraction via residual connections but is prone to overfitting on limited data; and LSTM is a lightweight model designed for sequential dependencies inherent in pulse waveforms. The customized CNN+LSTM architecture consists of a single convolutional layer with 32 filters, a kernel size of 5, and a ReLU activation function; three stacked LSTM layers with 64, 64, 32 hidden units respectively; and two separated dense layers for SBP and DBP prediction, each with 128 neurons. Each model has trained originally on MIMIC-III PPG-BP paired dataset, providing the pretrained weights [10][11]. To adapt the models to rPPG data, with all backbone layers frozen except for the final dense prediction layer, each pretrained model is fine-tuned on preprocessed rPPG–BP pairs described in the Data Analysis Section. For hyperparameter tuning, the Adam optimizer with initial learning rate as 1e-3 and mean squared error loss function as well as early stopping on validation loss with a default patience of 5 epochs, consistent with the methodology described in the paper that the pretrained models were trained on [4]. All models are trained on Google Colab T4 GPU with TensorFlow library. Performance is measured using MAE for SBP and DBP, refer to the Results section for details.

3.2 Blood Oxygen Methodology

Traditional SpO_2 measurement uses pulse oximeters and the Ratio-of-Ratios (ROR) method, while recent DL approaches achieve <2% RMSE, well above the 4% RMSE medical standard [12]. Three most notable works include: (1) An early but powerful work based on the ROR method that extracts Alternating Current (AC)/Direct Current (DC) components of the face from a ST-map of cropped facial video [13]. It then processes AC/DC in parallel with fused ResNet's to predict SpO_2; (2) A recent work that uses Eulerian Video Magnification with Hermite Transform (EVM-HT) to magnify colour changes of the face, and directly predicts SpO_2 through a 3D-CNN [14]; (3) ITSCAN which uses a Temporal-Shift Module Network with coordinate attention, instead of a 3D-CNN to efficiently handle spatio-temporal information [15]. The first and second works are computationally efficient, with the EVM-HT model showing the best accuracy. This project will explore the first method by Akamatsu et. al., which provides a simple, explainable, and efficient solution that can achieve <1.5% RMSE. Its explainability comes from the fact that it is based on the ROR method. The efficiency is due to the way the videos are compressed into single images (ST-maps) and allows for faster training on less memory.

There are 5 main parts to the architecture (Fig 9.3.1). (1) Face alignment, (2) ROI frames, (3) ST-map, (4) AC and DC component creation, (5) Parallel CNN model. The first 3 parts are discussed in the Data Analysis Section. In stage 4, the ST-map is passed through two filters, a low-pass and a band-pass. The low-pass filter extracts the baseline signal, taking low frequency components under 0.3 Hz, while the band-pass filter extracts mid-frequency components related to blood perfusion of 0.7 to

2.5 Hz. These are implemented with a finite impulse response filter and a 1D convolution for speed during training, and result in the DC and AC components. In stage 5 these components are processed through a pair of ResNet18 CNNs with intermediate fusion and late fusion. The intermediate fusion allows for information exchange between the models and uses a multi-modal transfer module (MMTM). Finally, the late fusion combines the outputs of the ResNet's using elementwise addition and produces the output through a fully connected linear layer.

ST-maps are scaled to [0,1], normalized using ResNet18 statistics, and paired with duplicated SpO_2 labels since each video provides only one measurement. Training uses a two-stage procedure: first freezing ResNet layers and training the fusion modules (LR = 1e-3), then fine-tuning the full network (ResNet LR = 1e-4, fusion LR = 1e-3). The model is optimized with Adam (weight decay 1e-5) and trained for roughly 20 epochs on a T4 GPU..

3.3 Body Mass Index Methodology

Like many other studies on facial-image-based body mass index prediction, our BMI measurement methodology follows a standardized pipeline consisting of data preprocessing, feature extraction, and prediction. Following preprocessing, which has been discussed in section 2.3, we employ transfer learning for feature extraction by adopting deep convolutional backbones. VGG19, providing a simple and stable architecture, and ResNet50, enabling deeper feature extraction, are employed and evaluated [16]. Both VGG19 and ResNet50 are initialized with ImageNet weights and tuned. To balance generalization with task-specific adaptation, only the deeper convolutional blocks (block 4, 5 in VGG19 and conv 4, 5 in ResNet50) are unfrozen while earlier layers remain fixed. This configuration enables the model to preserve broadly useful low-level visual features while allowing higher-level layers to specialize in identifying subtle facial cues correlated with BMI.

The extracted features are then passed into two types of prediction heads: 1) a residual-block-based regression head [8], intended to deepen nonlinear transformation of features and potentially capture subtle facial cues, and 2) a lightweight dense regression head, chosen to reduce overfitting risk and improve training stability in small datasets. See Fig. 9.4.1 in Appendix. Both combinations, ResNet50 or VGG19 paired with either regression head, output continuous BMI predictions, enabling systematic comparison of backbone capacity and head architecture within a controlled, consistent training framework. All models are trained on Google Colab T4 GPU with TensorFlow library.

4. Results

4.1 Blood Pressure

All three models improve noticeably after fine-tuning on the preprocessed rPPG dataset, confirming that transfer learning from PPG to rPPG tasks helps adapt to facial-video signals. Comparing the three model architectures, the 1D-CNN+LSTM model demonstrates the most stable training behaviour (refer to training and validation plots in Appendix) and achieves the MAE: 10.79 mmHg for SBP and 7.06 mmHg for DBP. The best SBP PCC is achieved by both 1D-ResNet and 1D-CNN+LSTM at 0.013. 1D-ResNet produces the best DBP PCC at 0.143. However, these PPC values are extremely low

since they are below what would be required to indicate meaningful correlation (should be close to 1.00) between true and predicted BP. Overall, no model reached a clinical standard on both SBP and DBP simultaneously, while still falling within reasonable error ranges as reported in the paper.

Method (Model)	Before Tuning		After Tuning			
	SBP MAE	DBP MAE	SBP MAE	DBP MAE	SBP PCC	DBP PCC
1D-AlexNet	12.85	11.64	10.86	7.15	-0.036	-0.041
1D-ResNet	18.49	7.27	**10.76**	**7.05**	**0.013**	**0.143**
1D-CNN+LSTM	15.54	6.99	**10.79**	**7.06**	**0.013**	0.069

Table 4.1.1. MAE results for each model architecture before and after fine-tuning, along with post-tuning PCC for both SBP and DBP.

	Original	Weighted	Oversampling
SBP PCC	0.013	-0.099	-0.020
DBP PCC	0.069	0.022	0.095

Table 4.1.2. PCC results for the original 1D-CNN+LSTM model, along with versions trained using weighted loss and oversampling of minority BP ranges. Additional analysis is presented in the BP Discussion section.

4.2 Blood Oxygen

Using two stage training with random crop and colour jitter resulted in the best training dynamics and RMSE of 1.48 and MAE of 1.01. This matches the results of the paper [17], but notably on a different dataset. However, the PCC is very low at 0.108. This poor correlation and generalization is visualized in Fig. 4.2.1. Although the results achieve the goal RMSE, the correlation coefficient is critically too low. Sample results showing successful and failed predictions are shown in Fig. 9.3.3.

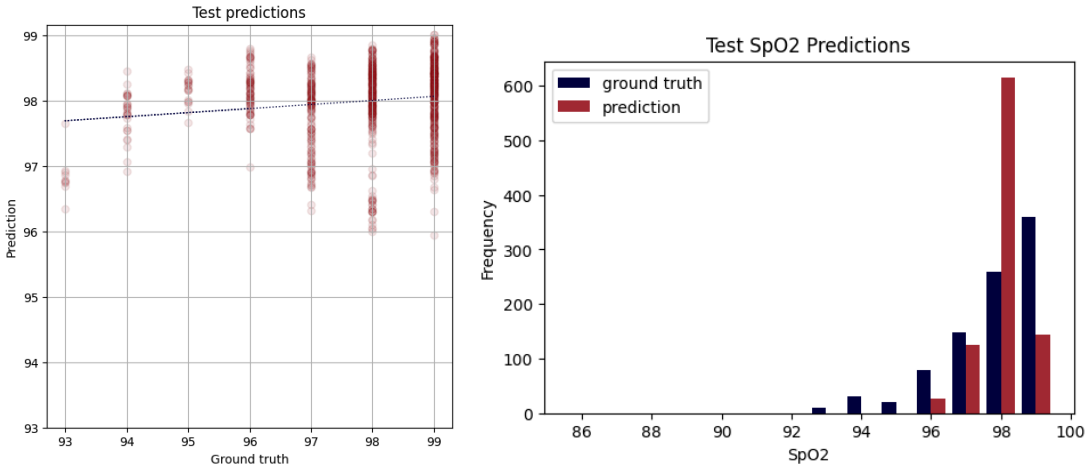


Fig. 4.2.1. Accuracy of trained model. (Left) Prediction vs. ground truth distribution as scatter plot. (Right) Prediction vs. ground truth as histogram. Predictions are binned to the nearest integer.

4.3 Body Mass Index

Compared with more complex tasks such as blood pressure, BMI estimation is employed with lower sensitivity to video noise and physiological variability. The best model, a tuned VGG19 combined with a dense head, achieved a low MAE of 2.47. This strong performance can be attributed to the architectural simplicity and stability of VGG19, which align well with the characteristics of our small dataset. In contrast, deeper architectures like ResNet50 or residual regression heads introduce additional complexity and are more prone to overfitting [16].

Method(Model)	MAE Before Tuning		MAE After Tuning	
	Residual Regression Head	Dense Head	Residual Regression Head	Dense Head
VGG19	2.8862	3.1514	2.7768	2.4661
ResNet50	3.2152	3.1431	2.9927	2.7466

Table 4.3.3. After tuning, the VGG19 with dense head got the best average MAE over the other combinations under 5-folds.

5. Discussion

5.1 Dataset Limitations

Overall, our project for predicting 3 biomarkers was able to meet most of the goal error rates, but not the correlation ratios for blood pressure and blood oxygen. One of the consistent issues is that of the dataset distribution and dataset quality. The skin tone bias towards lighter tones causes a higher error for darker skin tones which is noticeable in blood pressure estimation. Lighter skin inherently reflects green-channel intensity more effectively while reducing the signal quality in other channels for rPPG extraction. In terms of quality, the videos are high-bit rate and low resolution, however the face is small. A small face region affects blood pressure estimation noticeably as MTTS-CAN is sensitive to low-resolution skin patches where the small ROIs can amplify noise in the extracted BVP signal. Moreover, although all videos are captured in the same room and under similar lighting conditions, the space is not fully enclosed. As a result, recordings taken at different times can experience varying ambient light, which affects the light reflected from each individual's face, which is critical for reliable rPPG extraction. Furthermore, around one third of videos have flickering issues, which causes high frequency periodic dark shadows across the face. Although the flickering rate is fast, it still hinders the colour accuracy and interferes with prediction. This effect can be significant for both rPPG extraction in blood pressure prediction, and spatio-temporal map processing for blood oxygen prediction.

5.2 Blood Pressure

One of the requirements is to have the predicted BP error close to the medical-grade BP tolerance error. The results for BP are off by around 1 mmHg due to multiple data biases. However, returning to the client's goal of performing a holistic health assessment, evaluating overall quality of life across multiple dimensions, each biomarker prediction does not exceed high medical-grade precision standards. In terms of the model's architecture, 1D-CNN+LSTM generally outperforms 1D-AlexNet and 1D-ResNet by MAE and PPC because it simultaneously captures local pulse structure and long term temporal dependencies, such as beat to beat variations, which is essential for BP estimation. With pure CNNs, only spatial and short term temporal features can be extracted. Moreover, 1D-AlexNet and 1D-ResNet tend to overfit due to their large number of trainable parameters, despite applying weight

regularization or other overfitting mitigation strategies. However, 1D-ResNet tends to perform better than 1D-AlexNet because its residual blocks are able to capture detailed waveform features more effectively while preventing gradient degradation.

In summary, given the limitations of the current dataset, the 1D-CNN+LSTM model achieves the best with relatively low MAE due to its nature of combining local feature extraction with long term temporal modeling. One limitation of the 1D-CNN+LSTM model is its low PCC. Even after applying oversampling to underrepresented BP ranges (e.g., SBP < 100 or > 140, and DBP < 70 or > 80) and applying higher loss weights to these minority ranges during training, the overall PCC shows no meaningful improvement, as shown in Table 4.1.2 . This may be because these methods increase their influence during training but do not strengthen the correlations between the rPPG signal and true BP, so the model still cannot learn more related features. Going forward, the models need to be tuned and evaluated on the clients specific user dataset.

5.3 Blood Oxygen

Overall, this model produces RMSE and MAE errors within the goal, but the poor bland-altman plot and PCC reveals the shortcomings of this method and the dataset. Several techniques were used to try to combat the flaws of the dataset. First, a weighted random sampler was used to increase the chance the model saw the rarer blood oxygen values. This resulted in a smoother training and validation loss curve, but ultimately did not improve performance greatly. Second, even with the use of data augmentation methods, the results did not improve significantly. Adding colour jitter and random crop reduced the training-validation gap, but was not enough to improve the PCC.

Another limitation of this method is using the pretrained ResNet on ST-maps. Since ResNet is pretrained on image net and ST-maps are temporal images, there is a domain gap which has to be learned. This was addressed in the original paper [11], and still proves to improve the error and generalization of the model. Compared to existing methods, it performs similarly to methods such as [13][17], but is much worse than the EVM-HT 3D-CNN method by Andres [12]. This more sophisticated model is worth trying in the future to improve SpO_2 prediction. Interestingly, these models share a lot of similarity in their high-level design. The EVM-HT 3D-CNN method effectively strengthens the AC components of the video through colour magnification and is added on top of the original video. This enhanced video is processed using a more expressive 3D-CNN. Overall, the combined effects of the data distribution, data quality, and method limitations are the causes for shortcomings in PCC and generalization performance.

Method (Dataset)	RMSE	MAE	PCC
Ours (MCD-rppg)	1.485	1.01	0.108
Akamatsu [11] (personal)	1.40	1.18	0.496
ITSCAN [13] (VIPL-HR)	1.19	1.10	--
Andres [12] (personal)	0.532	0.375	0.962

Table 5.3.1. Evaluation of DL-based models on facial SpO_2 prediction.

5.4 Body mass index

Because BMI is not measured against a strict clinical standard and is instead treated as an approximate indicator of body fat, the objective of our model is to produce predictions that are as close as possible to the ground-truth labels used in the dataset with the MAE evaluation. Although our results are not as strong as those reported in some prior works [8][16], the performance, consistent with our observations in blood pressure and SpO_2 tasks, is fundamentally constrained by the size and distribution of the dataset. The limited number of subjects and imbalance across BMI groups reduces the model's ability to generalize to underrepresented categories. We attempted to address this by applying class weights to compensate for the unbalanced distribution, but this adjustment led to worse performance, see table 9.4.3 in Appendix, likely because the model overcorrected toward minority groups and destabilized the regression learning process. Despite these limitations, the model still demonstrates reasonable performance given the constraints, showing that a lightweight architecture like VGG19+dense remains effective for small-scale physiological estimation tasks.

5.5 Ethics

We use a public dataset which is available on hugging face for download. During deployment on the **** platform, there are concerns related to privacy, fairness, and risk of misinterpretation. Improving and fine-tuning our models will be done in the future following ****'s policies. Importantly, **** is committed to privacy and is certified for GDPR, HIPAA, among others [18]. Critically, the purpose of personalized healthcare is not to substitute doctors, but rather to inform and educate the user.

6. Implementation

We upload all code, processed datasets, trained models, and configuration files to the shared folder provided by ****. These materials include complete pipelines for BP, SpO_2, and BMI, along with documentation so the client can run and test the models on their own data. All three biomarkers are designed to be predicted in parallel from the same Face Scan video, minimizing any additional burden on ****'s existing workflow.

The models have not yet been deployed by the client; all experiments were conducted offline using public datasets. The next key step is that all models should be fine-tuned on the client's private user dataset to reduce bias, improve generalization, and match the distribution of real-world **** users. The architecture also supports user-specific fine-tuning, where the client can adapt the last layers of the model to individual users or user cohorts to improve long term prediction accuracy. Additional implementation considerations include selecting an appropriate deployment strategy (cloud vs. on-device), ensuring fairness across demographic groups, and performing further validation on client data before production use. The provided codebase and model checkpoints offer a ready foundation for **** to begin integration and fine-tuning.

7. Conclusions & Future Directions

This project explored the feasibility of extending ****'s Face Scan feature to estimate blood pressure, blood oxygen saturation, and body mass index from facial video using DL. Across all three pipelines,

we successfully built complete end-to-end systems compatible with ****'s existing MTTS-CAN rPPG workflow. Our results partially met the original goals: BMI prediction performed well (MAE ≈ 2.47), SpO_2 met the target RMSE (<1.5) though with poor correlation, and BP achieved reasonable errors (SBP ≈ 10.79 mmHg, DBP ≈ 7.05 mmHg) but lacked strong predictive correlation. These findings indicate that contactless biomarker estimation is feasible but remains limited when trained on public datasets such as MCD-rPPG.

The main barriers were the limited dataset size, demographic imbalance, and video quality issues. With only one label per subject and a strong skew toward lighter skin tones and young adults, the models lacked the diversity needed for strong generalization. Small face regions, lighting flicker, and inconsistent frame rates further reduced rPPG signal quality and caused deeper architectures to overfit. Despite these constraints, the project provides **** with clear, practical value: a complete codebase, trained models, and modular pipelines that can be fine-tuned on the client's private dataset to improve performance and fairness. The findings also highlight promising modeling directions, CNN+LSTM for BP, Akamatsu-style ST-maps for SpO_2, and VGG19+dense for BMI, and areas requiring better data and improved architectures.

For future development, we recommend collecting more diverse paired video–biomarker data, exploring more advanced temporal architectures (e.g., 3D-CNNs, transformer-based models, or EVM-HT variants), and performing subgroup validation to ensure fairness across skin tones and age groups. Beyond these, exploring multi-task learning to jointly predict multiple biomarkers, user-specific fine-tuning to improve long-term accuracy for frequent users, and real-world testing on naturalistic smartphone videos will be important next steps. With improved data and continued refinement, these methods have strong potential to enhance ****'s holistic, contactless health assessment capabilities.

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[18] ****, *“**** AI’s Radical Privacy Policy,”* ****, https://****/radical-privacy (accessed Dec. 1, 2025).

9. Appendix

9.1 Data Analysis

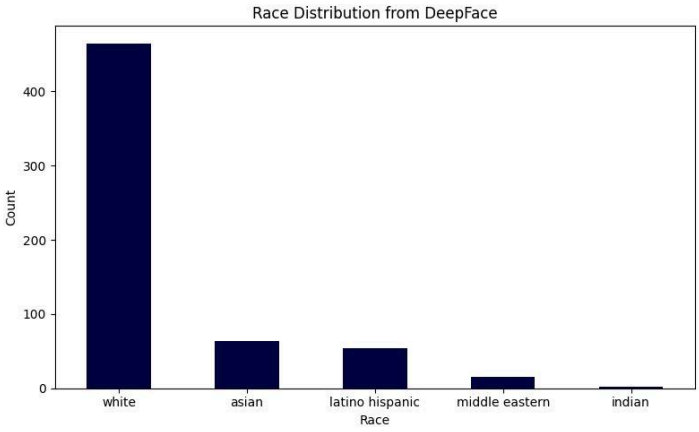


Figure 9.1.1. White participants dominate with around 450 out of 600 participants.

9.2 Blood Pressure

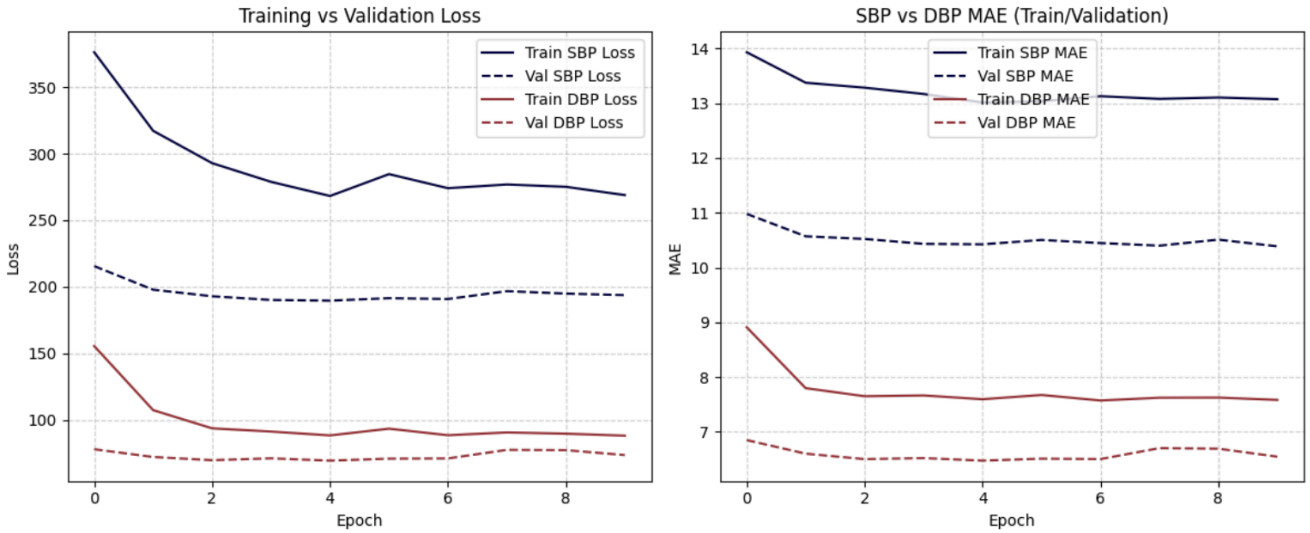


Figure 9.2.1. 1D-AlexNet fine-tuning training and validation curves.

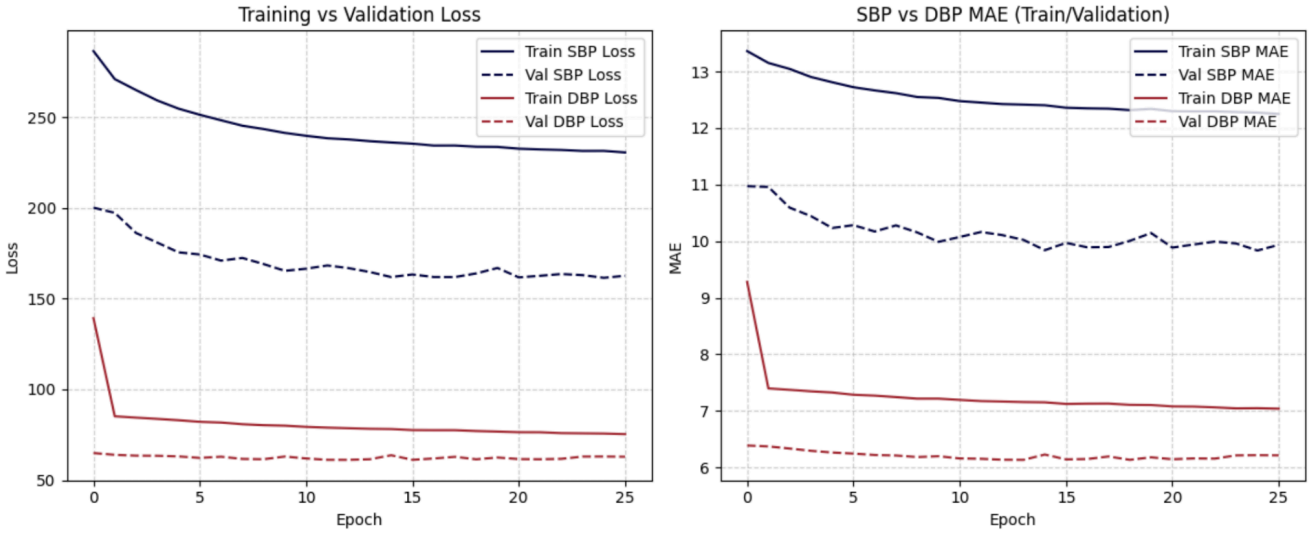


Figure 9.2.2. 1D-ResNet fine-tuning training and validation curves.

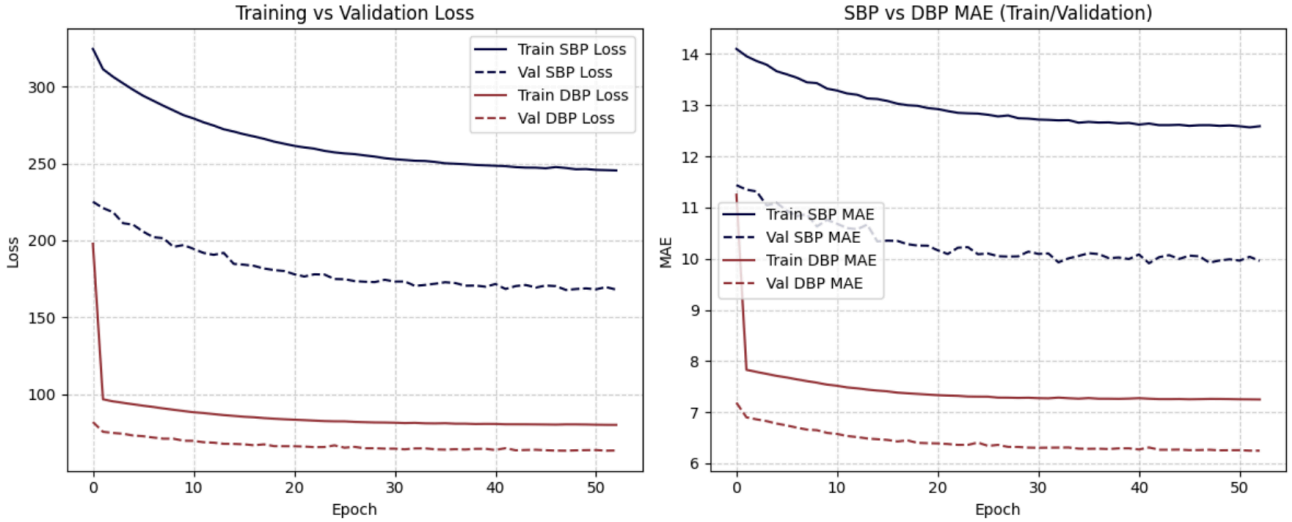


Figure 9.2.3. 1D-CNN+LSTM fine-tuning training and validation curves.

A sample prediction for BP:

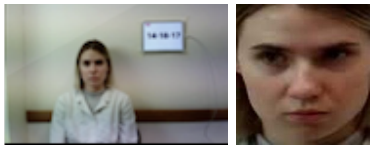


Figure 9.2.4. First, extract the face from each frame of the full video.

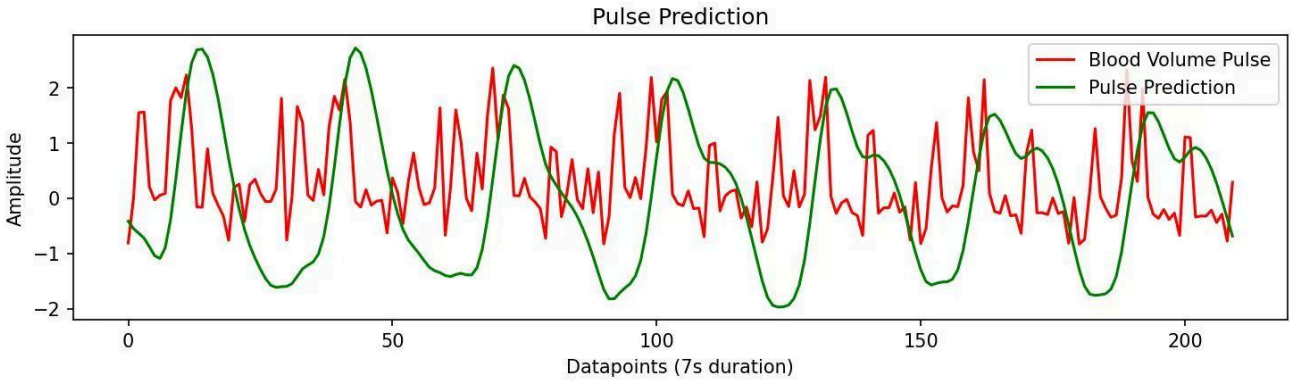


Figure 9.2.5. Then, sample processed window (30 Hz) segments from this video.

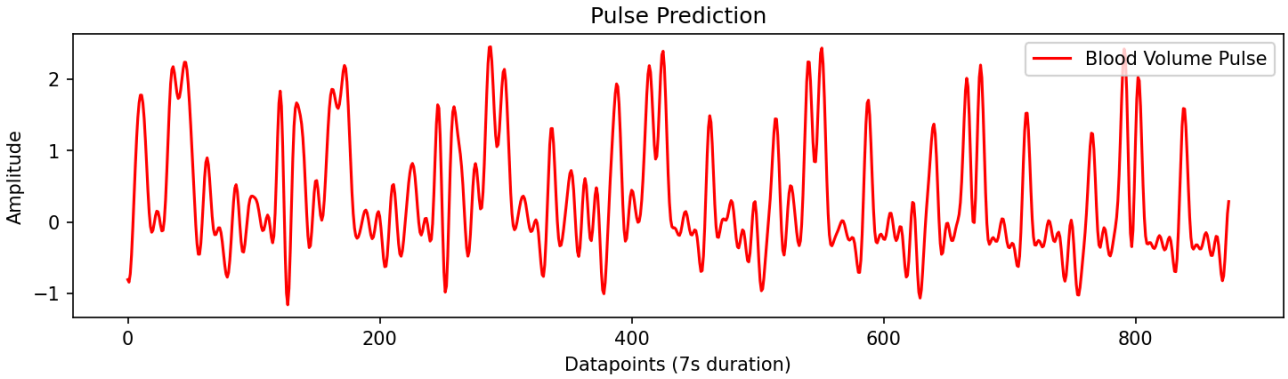


Figure 9.2.6. Upsampled to a 125 Hz sampling rate.

The model predicted an average (across all segments from this video) of 116.6 mmHg for SBP and 73.0 mmHg for DBP, compared with ground-truth values of 105 mmHg and 78.0 mmHg, respectively.

9.3 Blood Oxygen

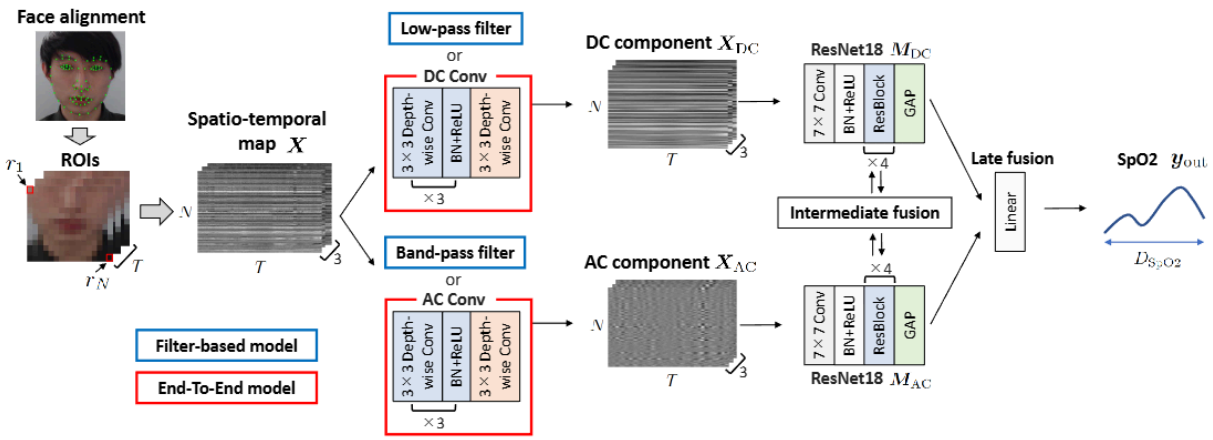


Fig. 9.3.1. Overview of the method. From [11]

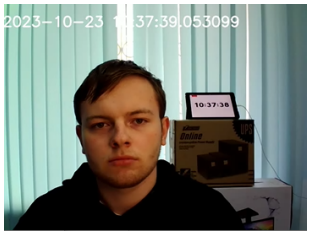
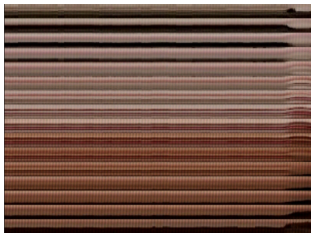
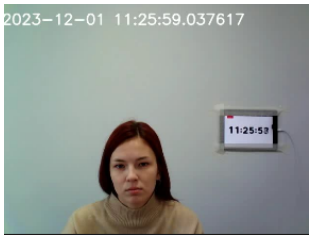
Video (one frame)	Spatio-temporal map	Prediction (SpO_2 %)	Ground Truth (SpO_2 %)
		97.9	97
		98.08	99
		98.72	99

Fig. 9.3.2. Randomly sampled predictions using the Akamatsu architecture.

Training loss curve and comparison of prediction vs. ground truth SpO_2 for one fold.

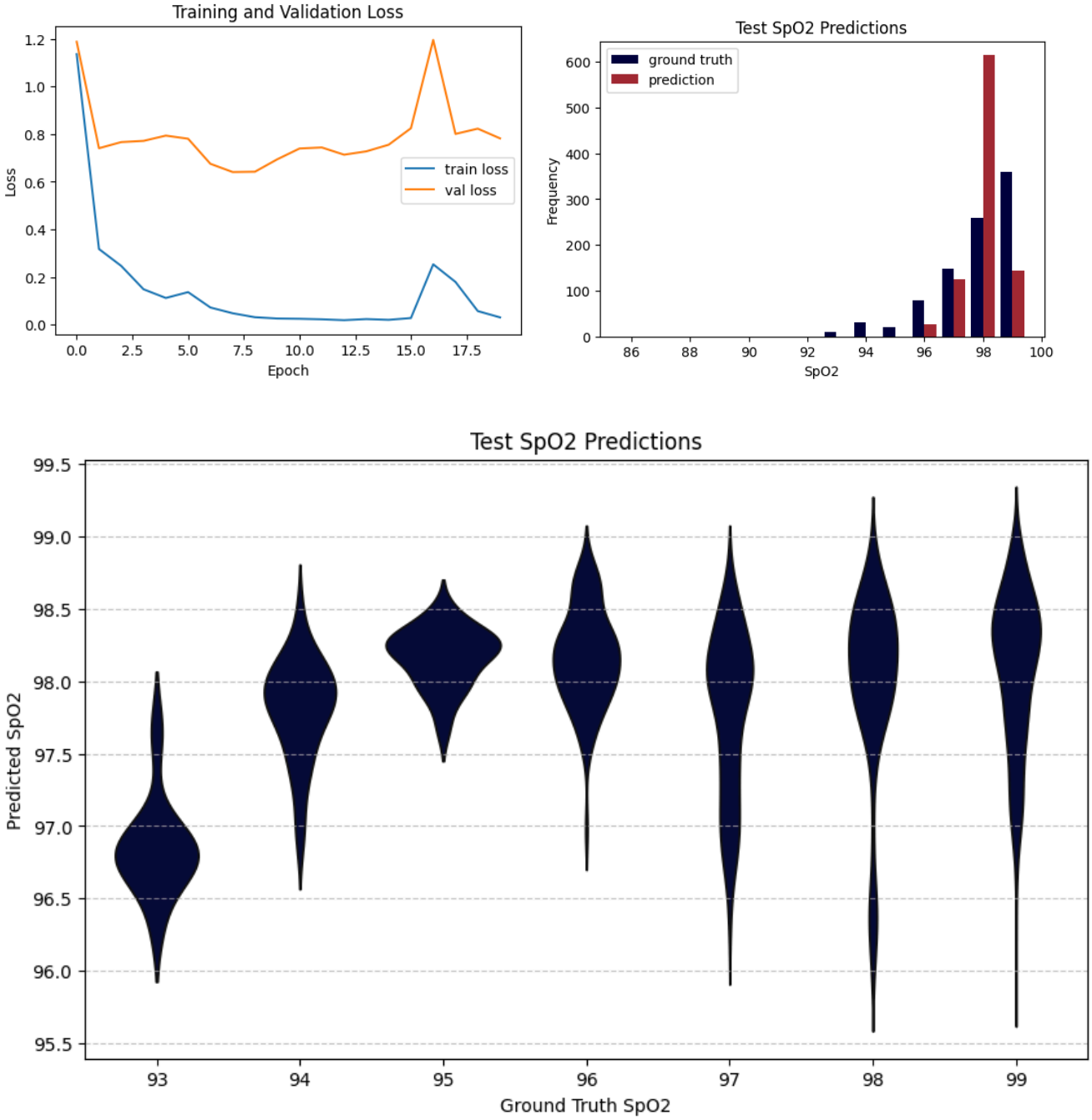


Fig. 9.3.3: a) (Top left) Training and validation loss curve for one fold of cross-validation. Loss is measured as mean squared error (MSE). b) (Top right) Histogram comparing ground truth and predicted blood oxygen. The predicted values are binned to the integer with range of [-0.5, 0.5] from the ground truth. c) (Bottom) Violin plot comparing predictions and ground truth showing distribution of guesses. Note that violin plots are scaled to fit a constant width.

9.4 BMI

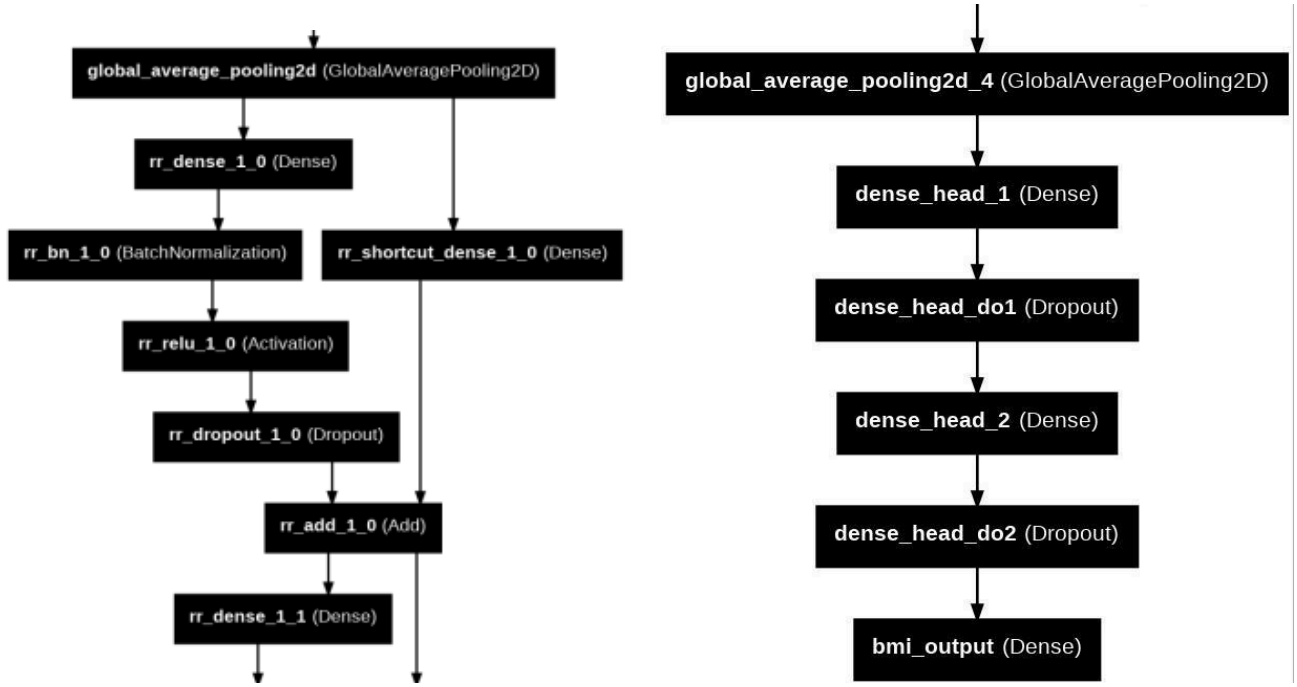


Figure 9.4.1: (1) Left side: residual-block-based regression head, in this experiment we put 2 blocks × 3 sub-blocks which shows the best performance in study of Pham et al.[8], (2) Right side: A lighter-weight dense regression head



Figure 9.4.2: Sample BMI predictions from MTCNN-cropped faces.

VGG19+Dense Fold 3	Underweight (71 samples) BMI < 18.5	Normal (384 samples) 18.5 ≤ BMI < 25	Overweight (103 samples) 25 ≤ BMI < 30	Obese (42 samples) BMI ≥ 30
No Weighted	2.1053	1.2953	2.6839	5.5685
Weighted	2.0420	1.5315	3.0748	9.6209

Table 9.4.3. After giving larger weights for samples in less frequent groups, the overall MAE scores got worse especially for the least frequent group, obese.

Attribution Table

Team: Project Group 2, Project D - ****

Deliverable (circle): proposal / project update / final presentation / **final report**

Contribution	**Student Names:**			
	Liza	**Samson**	**Jiale**	**Ziyi**
Background	RD, FD	RE	RE	RE
Data Analysis	FD	RD, RE	RE	RE
Methods	RD, FD, RE	RD, FD, RE	RD, FD, RE	RD, FD, RE
Results	RD, FD, RE	RD, FD, RE	RD, FD, RE	RD, FD, RE
Discussion	IR, RE	BS, IR, RD, FD	BS, IR, RD,FD	BS, IR, RD, FD
Implementation	RD, RE	RE	RE	RE, FD
Conclusion	RE	RE	RE	FD
Blood Pressure	C/DB, T	DB, T	DB, T	SD, C, DB, T
SpO_2	C/DB, T	SD, C, DB, T	DB, T	DB, T
BMI	C/DB, T	DB, T	SD, C, DB, T	DB, T

TD: Team Discussion

CD: Client Discussion

RD: Rough Draft

SD: Solution Design

C: Coding

DB: Debugging

T: Testing

BS: Brainstorming

IR: Independent Research

FD: Final Draft

RE: Revision

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_____Ziyi Huang_____
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_____Jiale Li_____
Name: Jiale Li